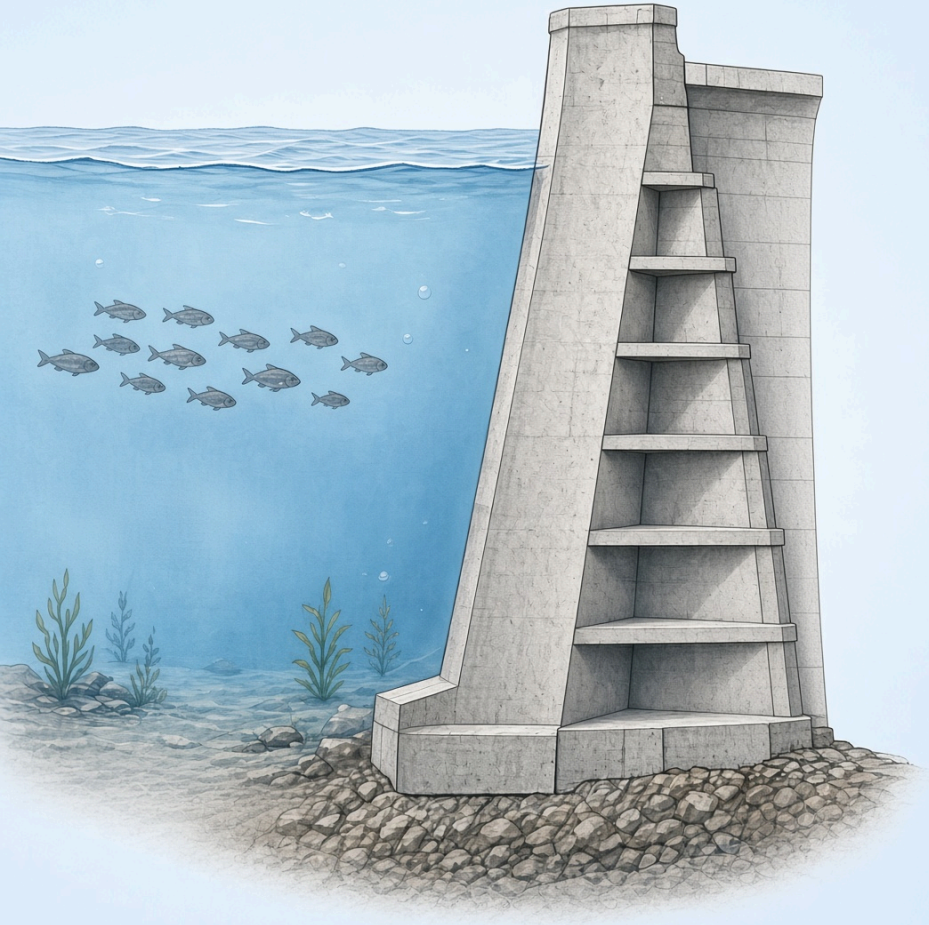


Parcie hydrostatyczne na powierzchnie zakrzywione

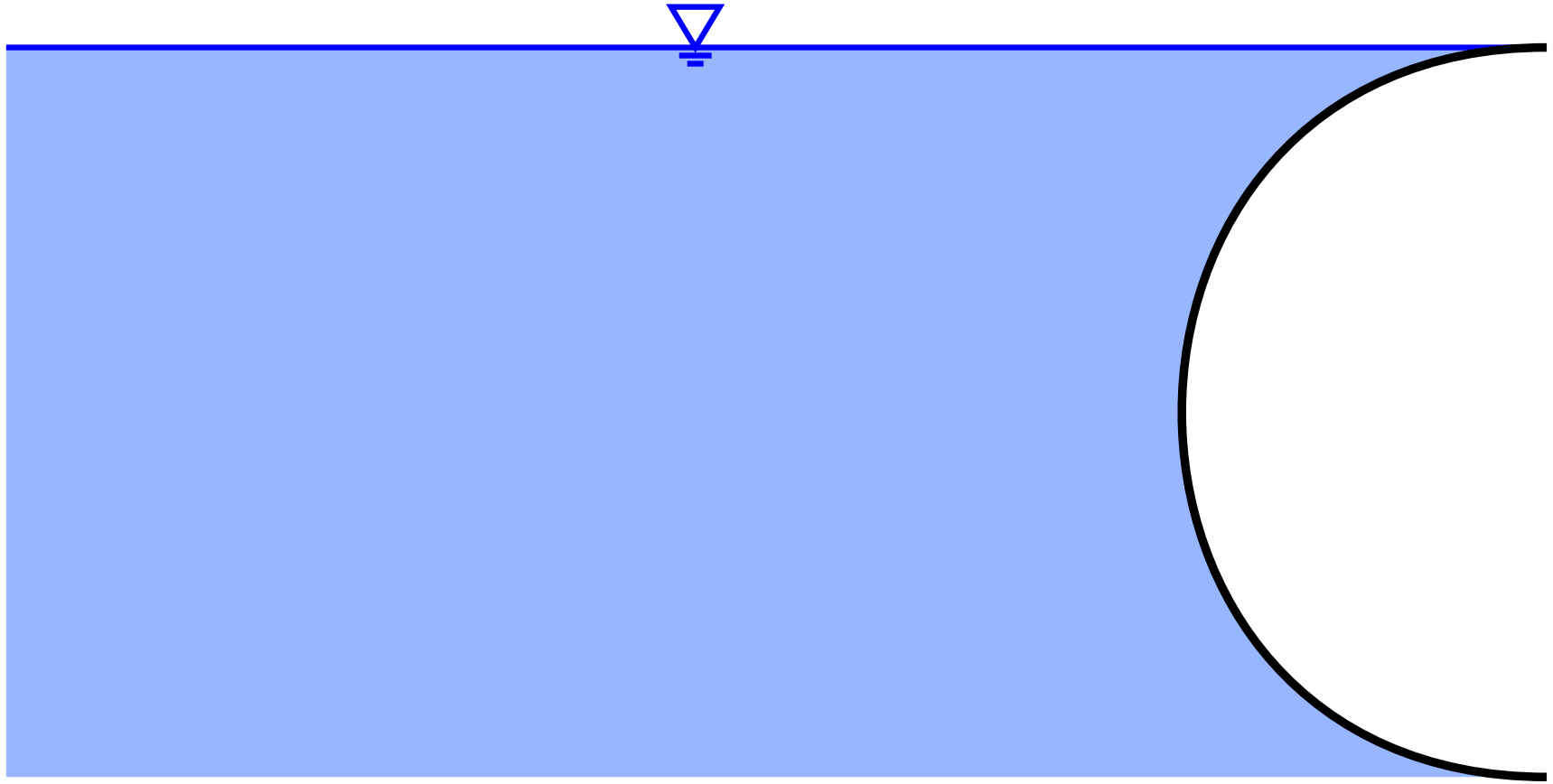
Adam Kiczko

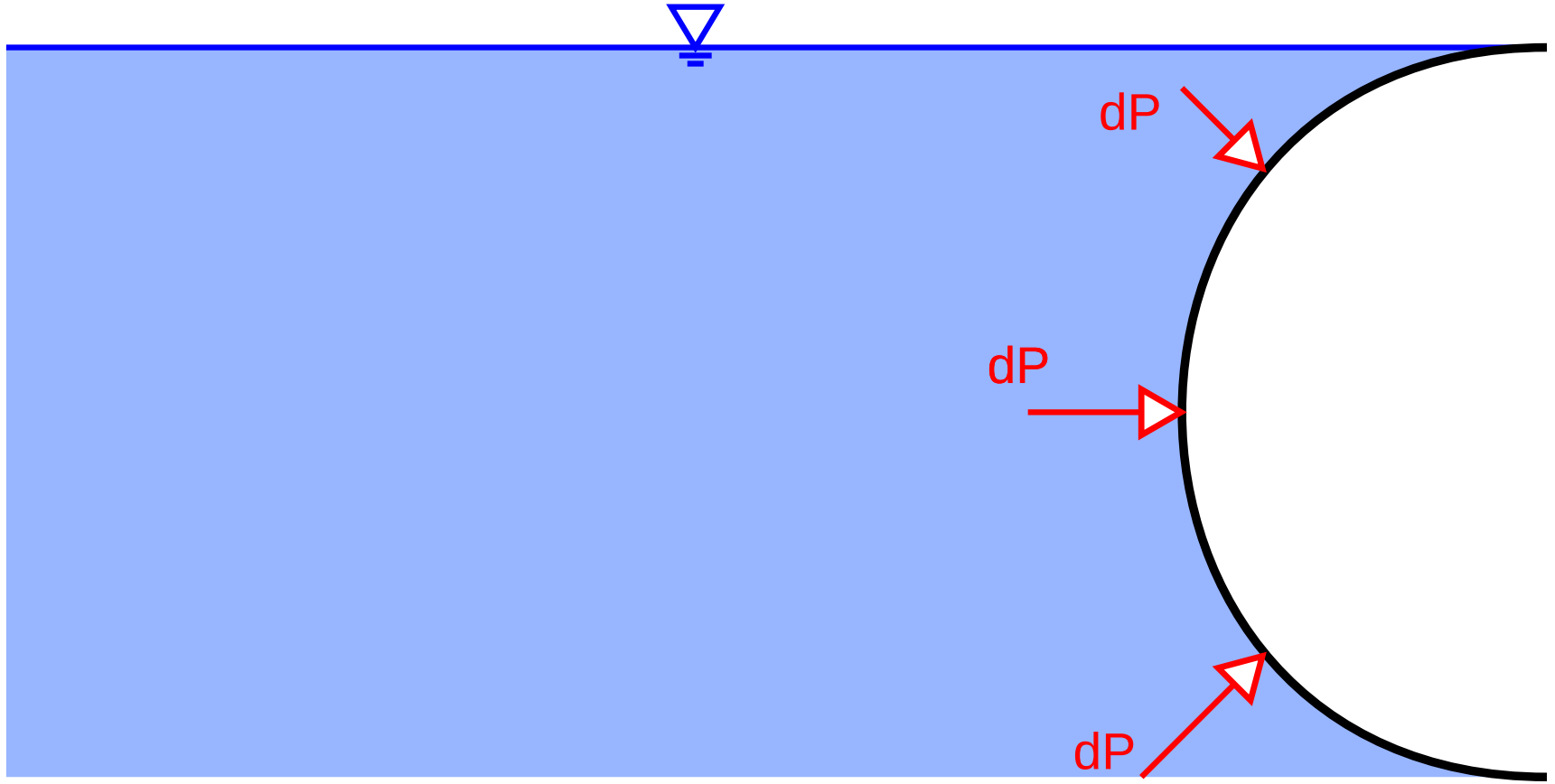
adam_kiczko@sggw.edu.pl

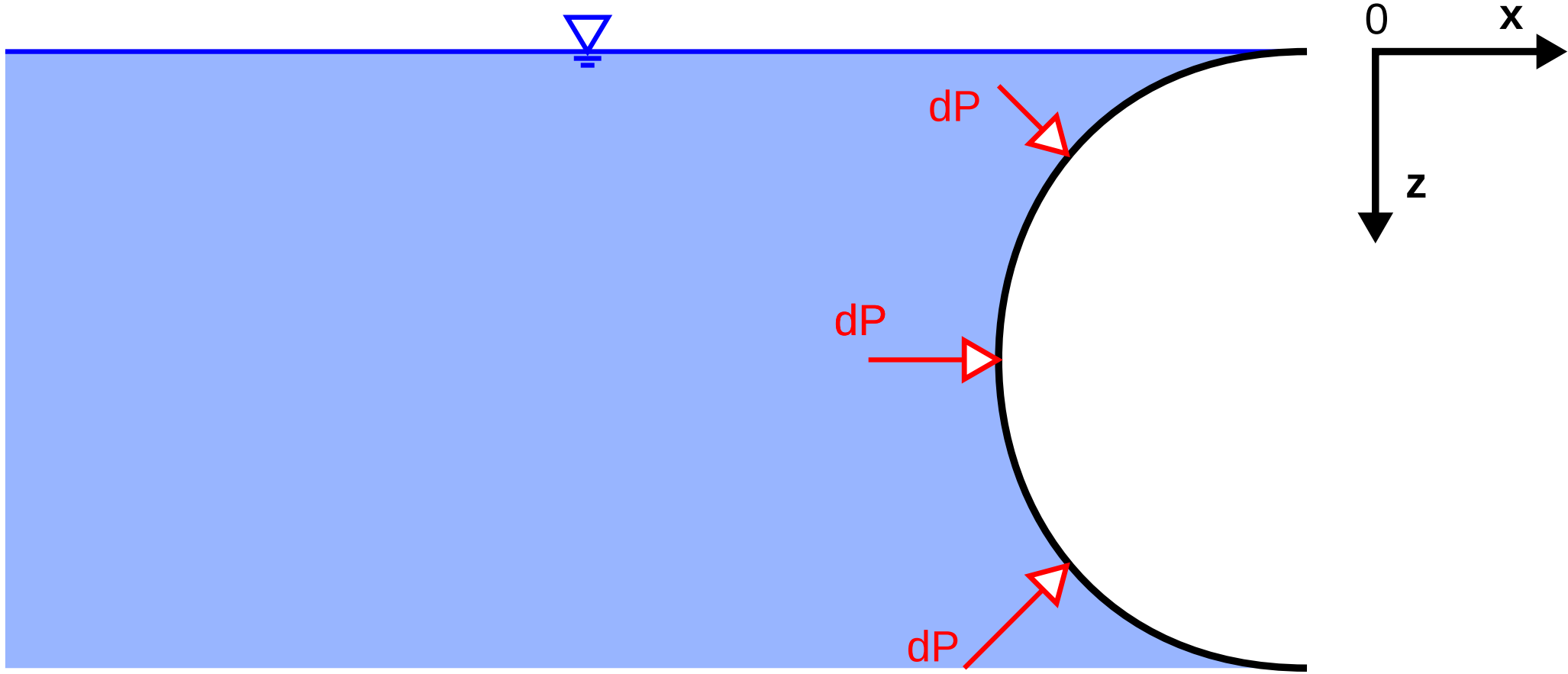
Plan wykładu

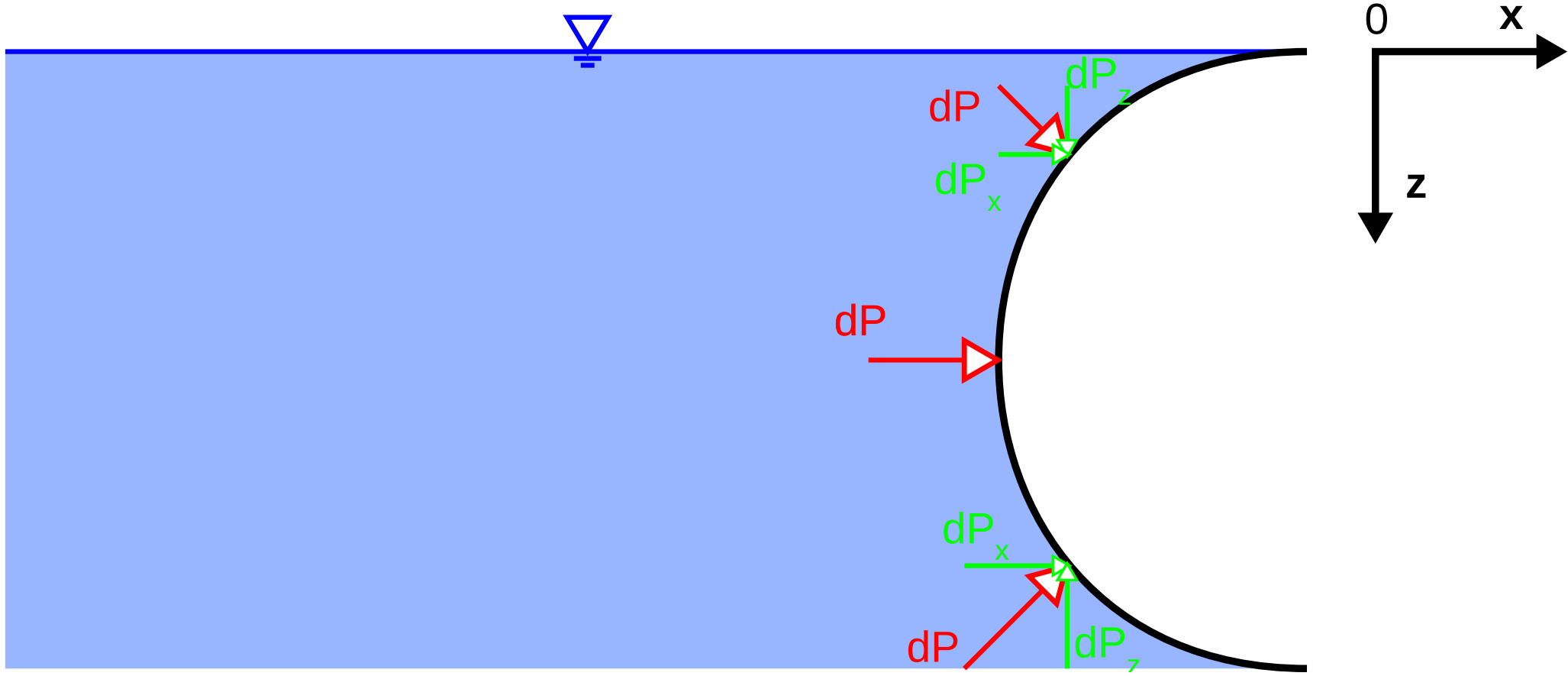
1. Parcia na powierzchnie zakrzywione
2. Przykłady
3. Wypór hydrostatyczny

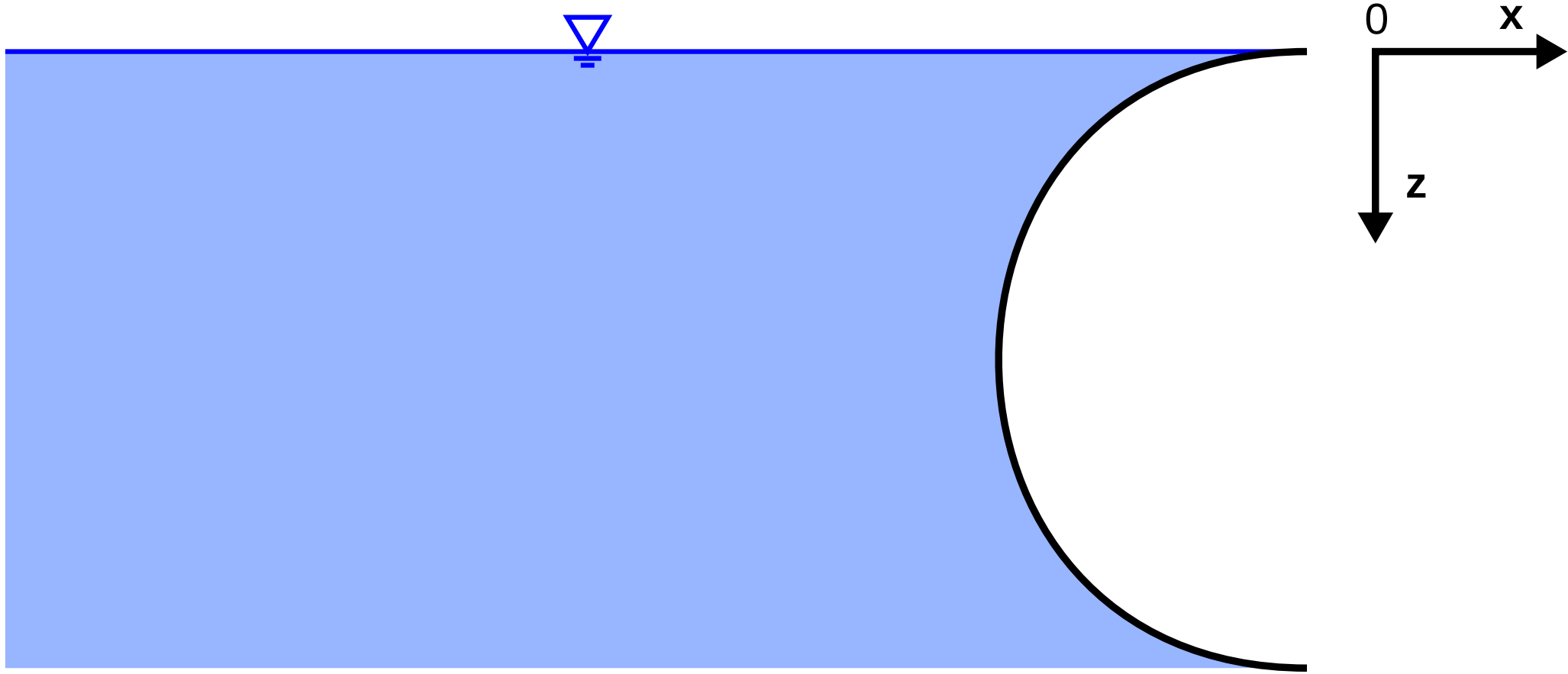


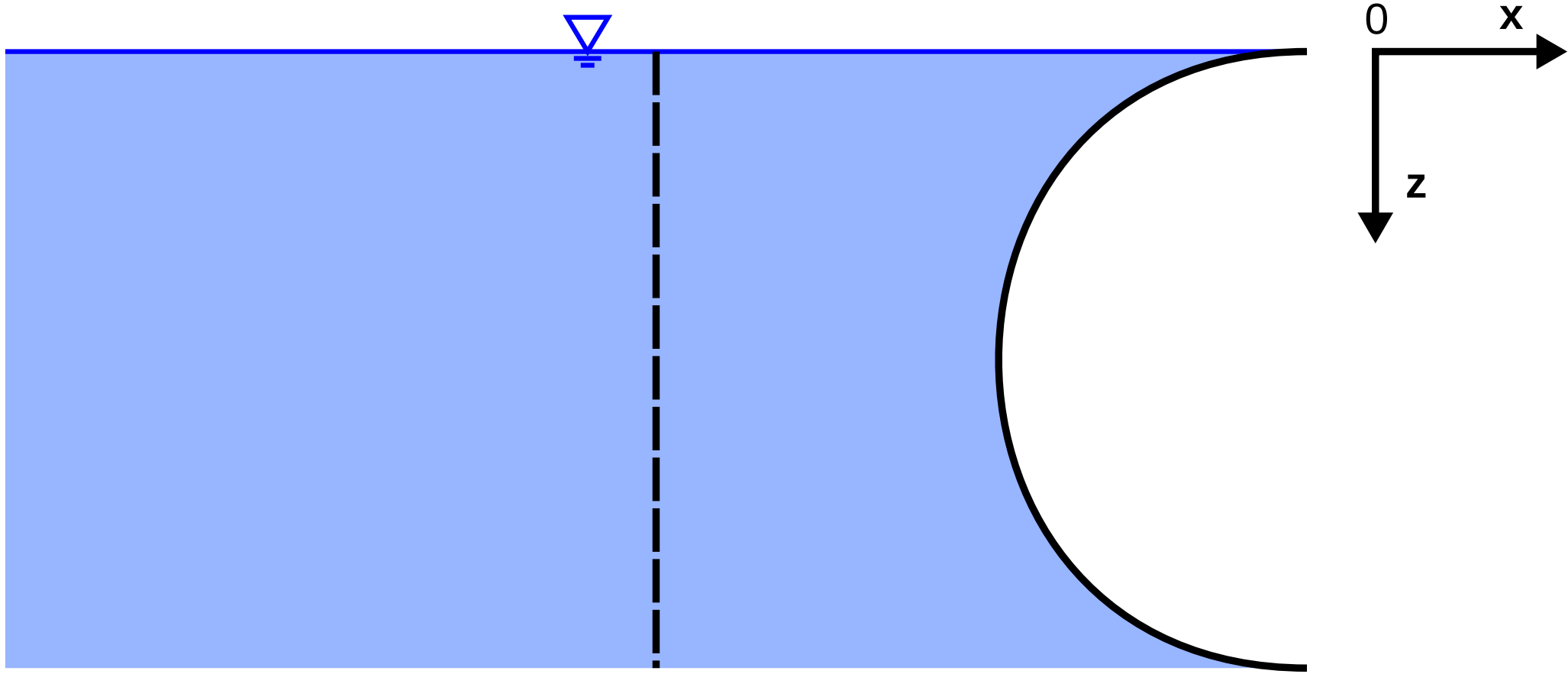


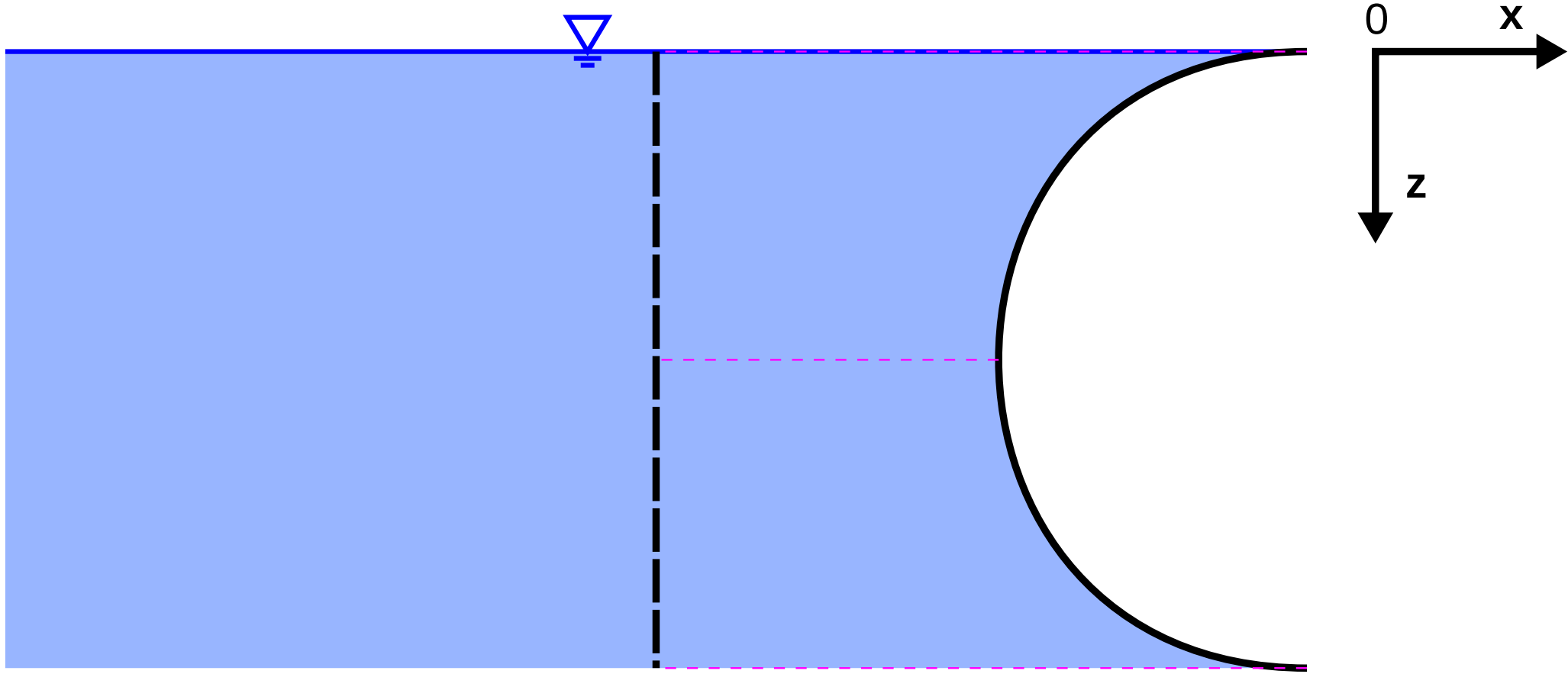

dP
dP
dP

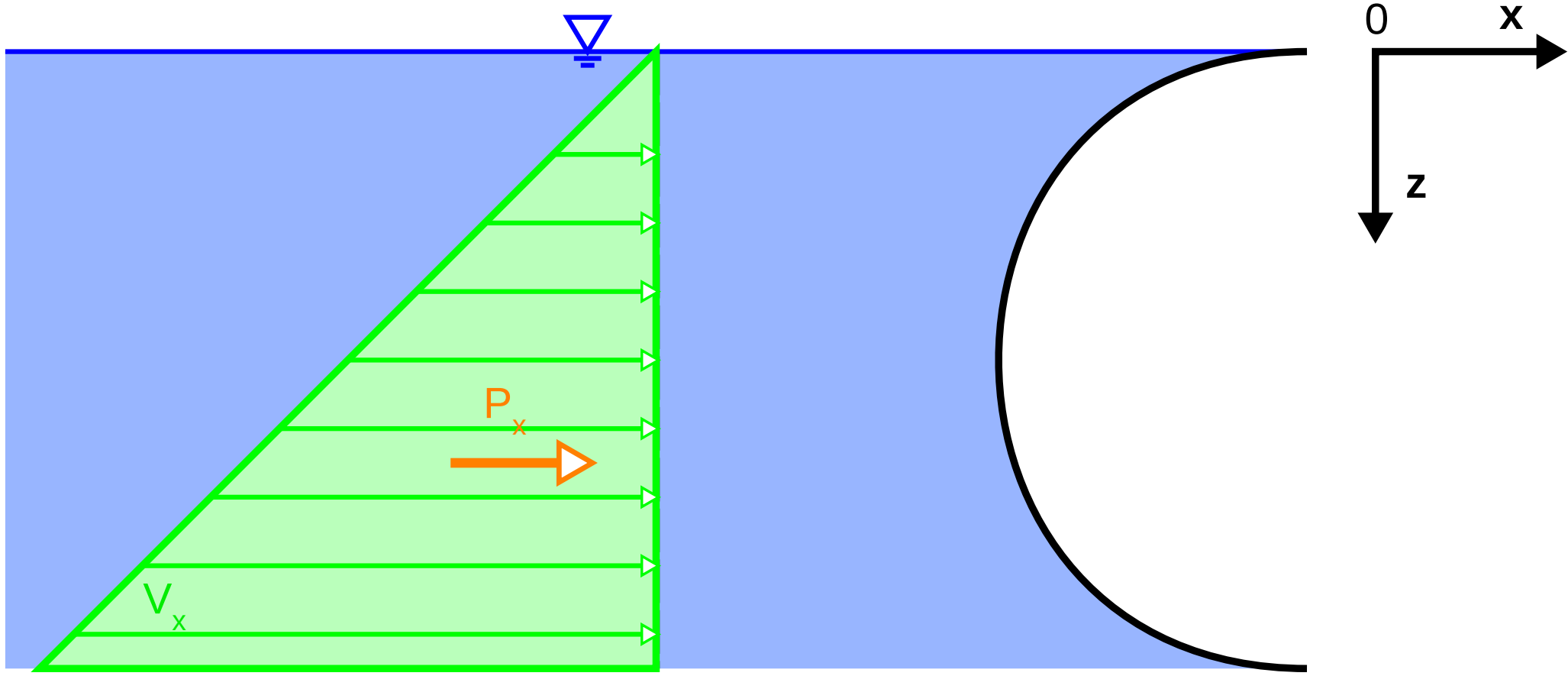

0
x
z
dP
dP
dP

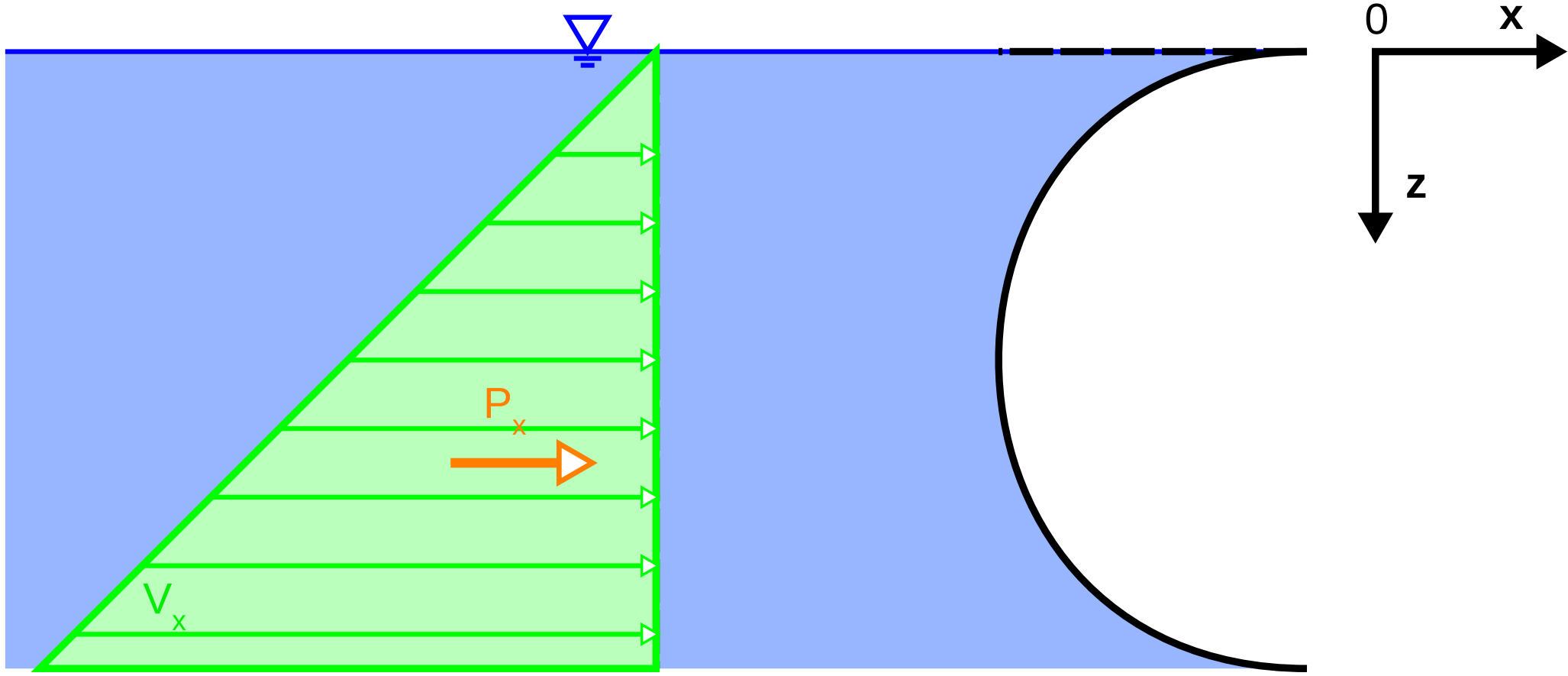

0
x
z
dP
dP
z
dP
x
dP
dP
x
dP
dP
z

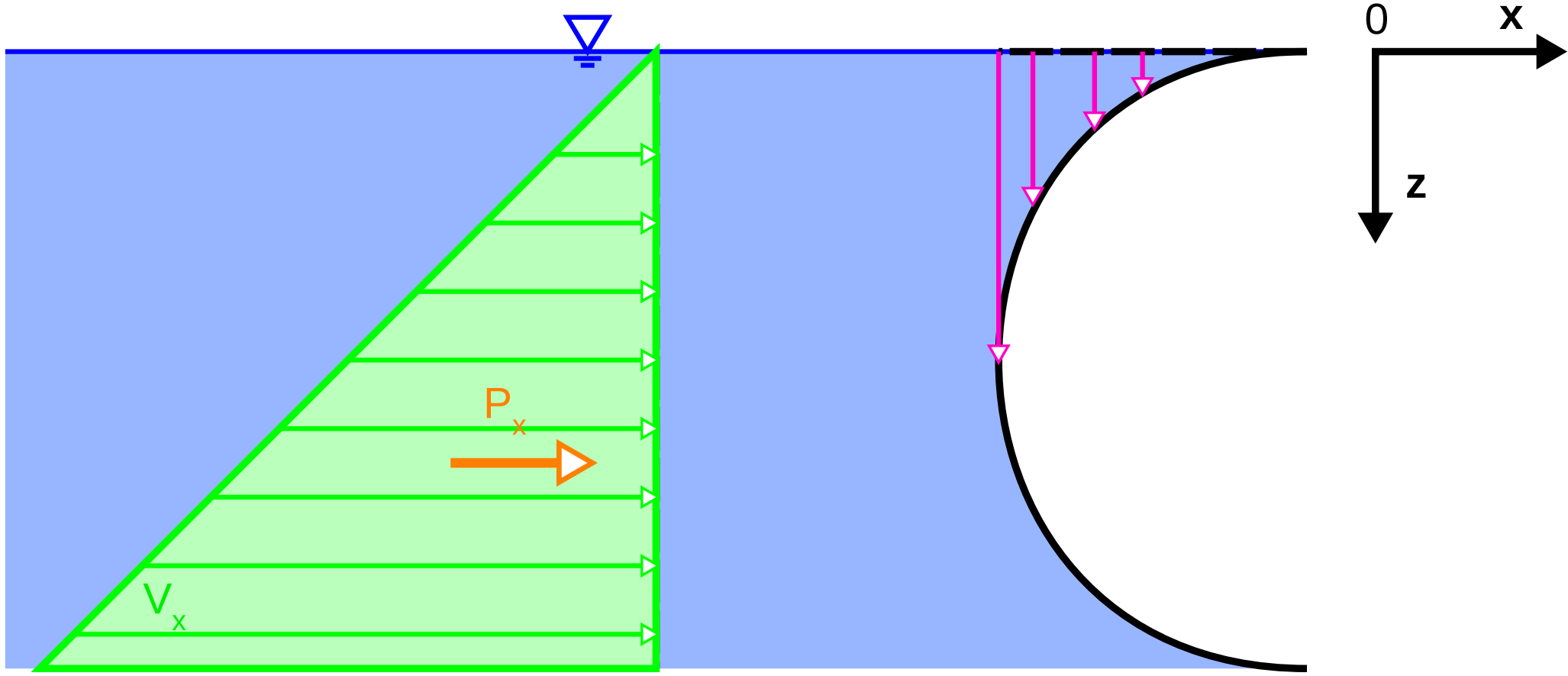

0
x
z

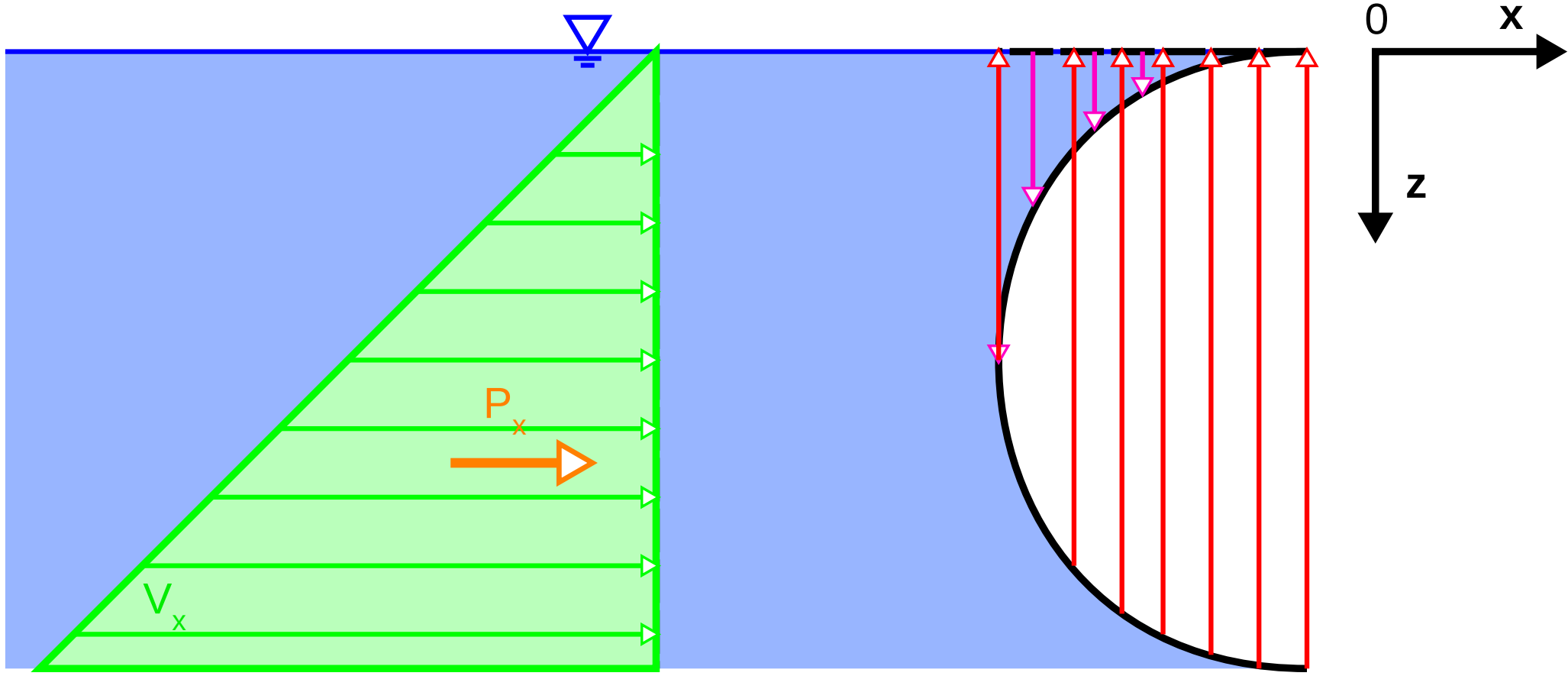

0
x
z

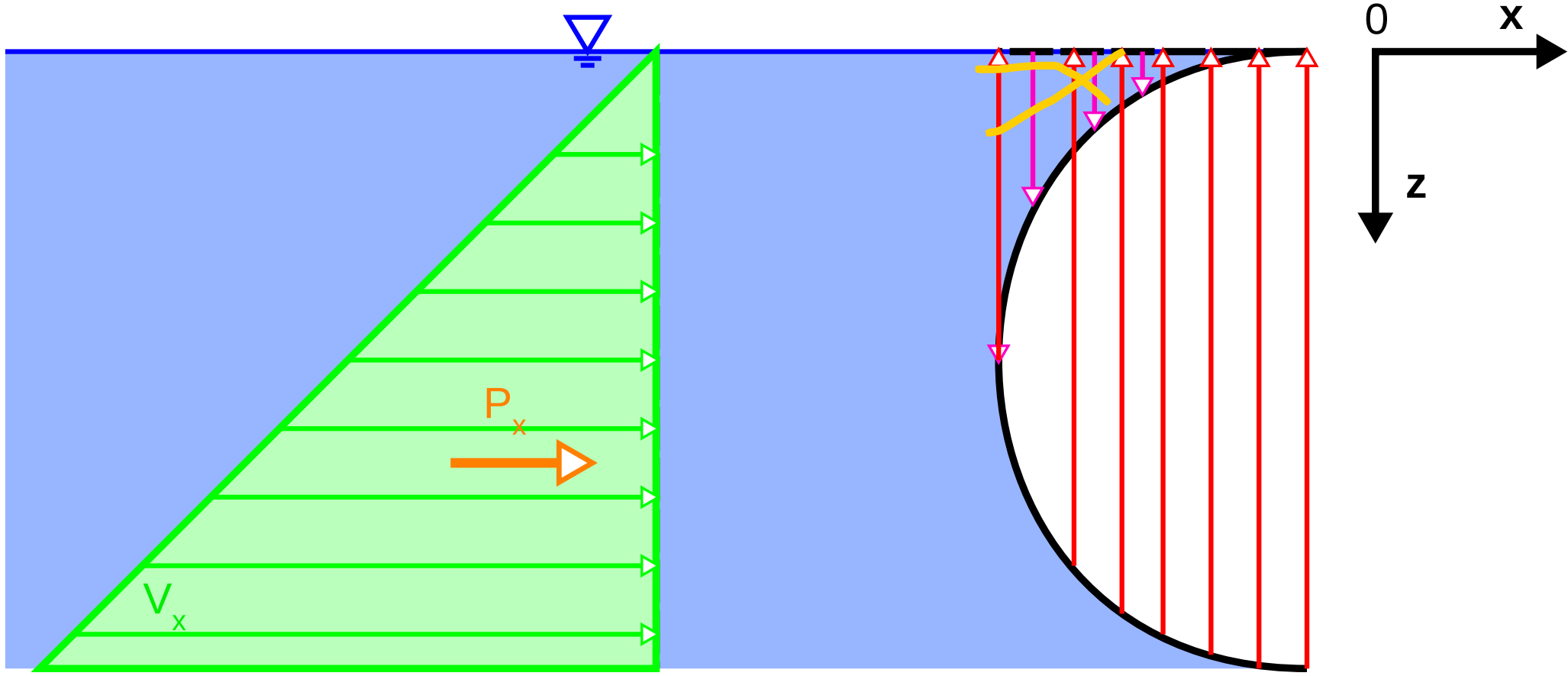

0
x
z

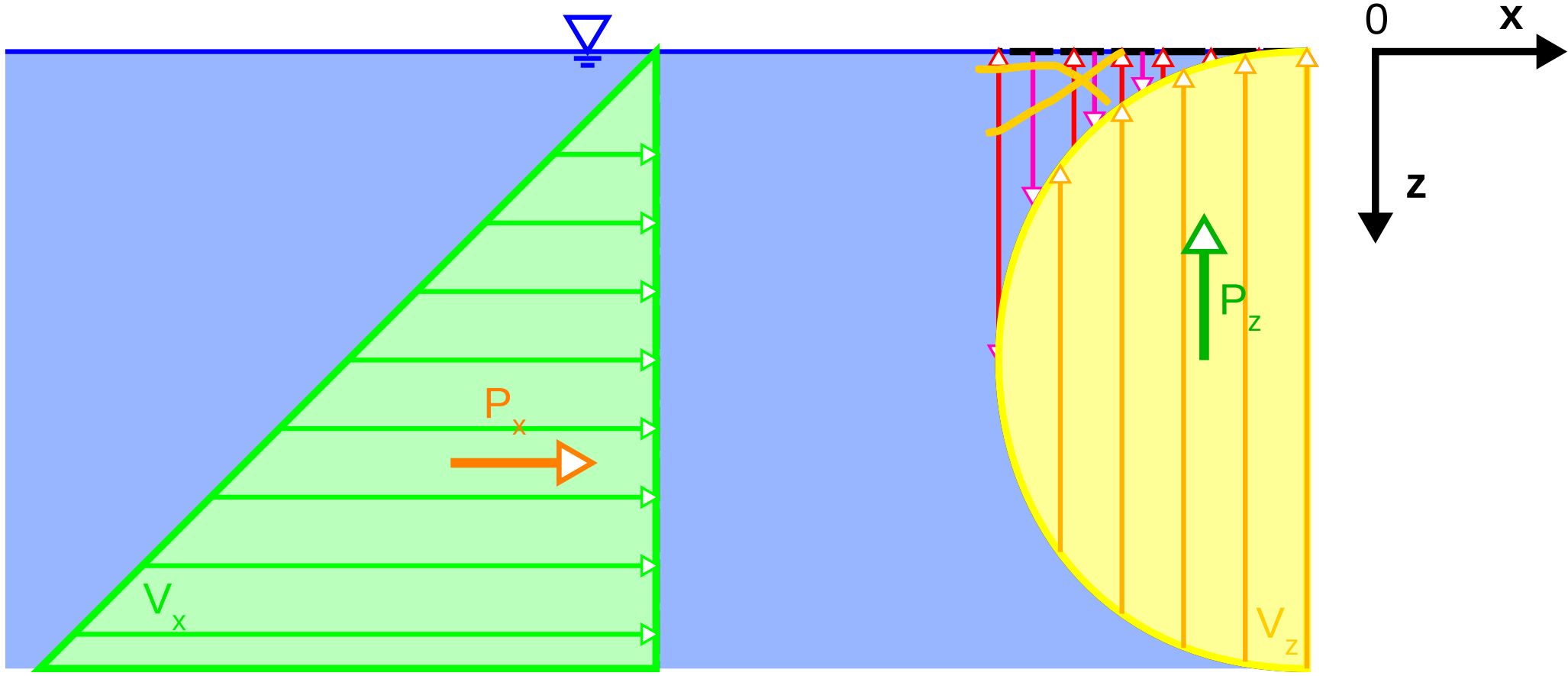

0
x
z
P_x
V_x

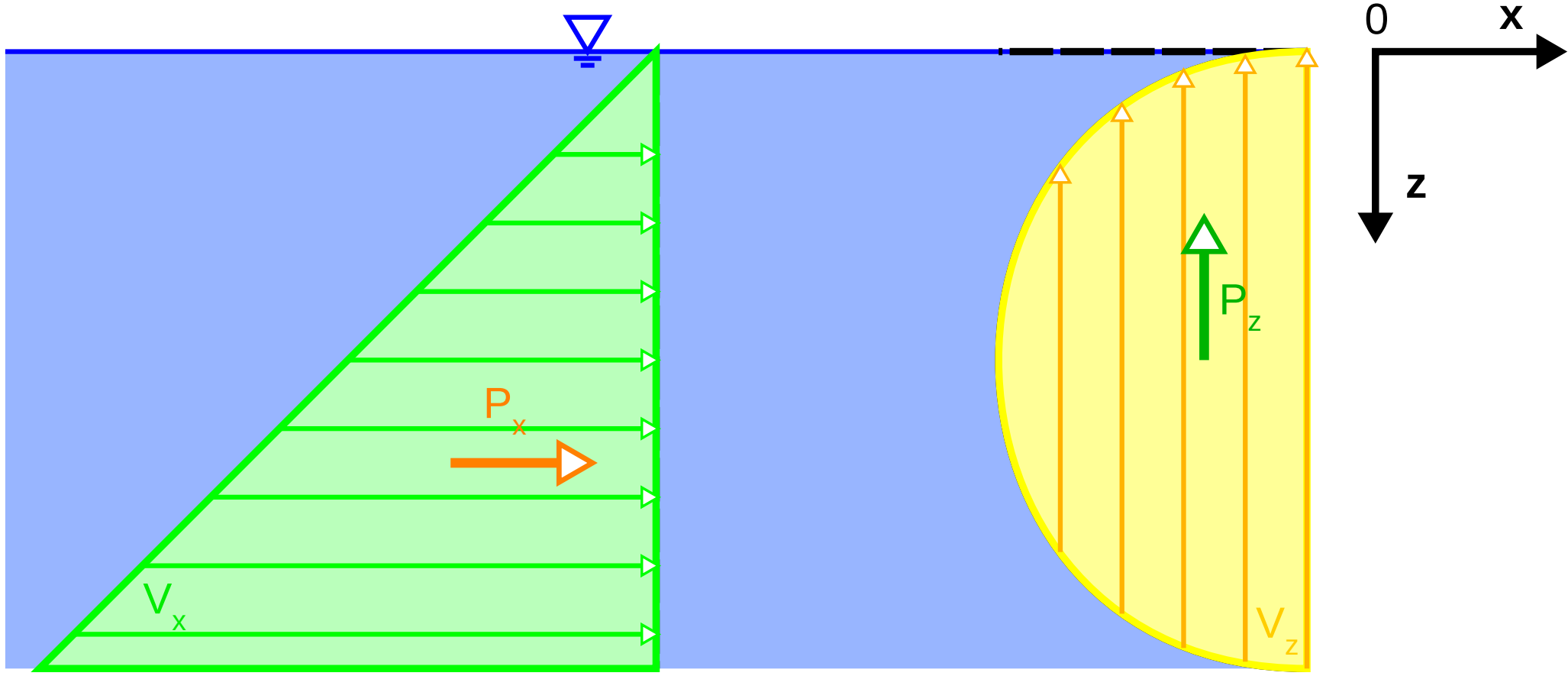

0
x
z
P_x
V_x

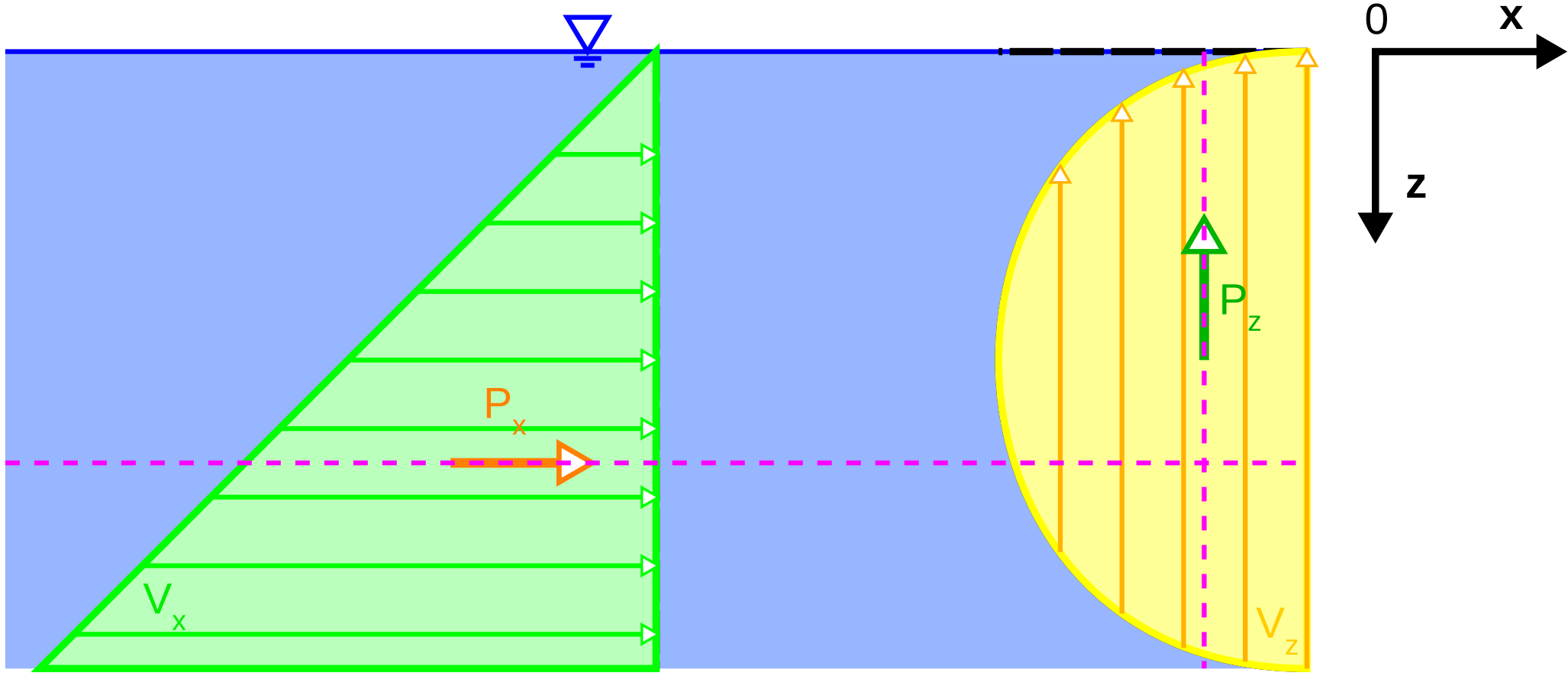

0
x
z
P_x
V_x


0
x
z
P_x
V_x

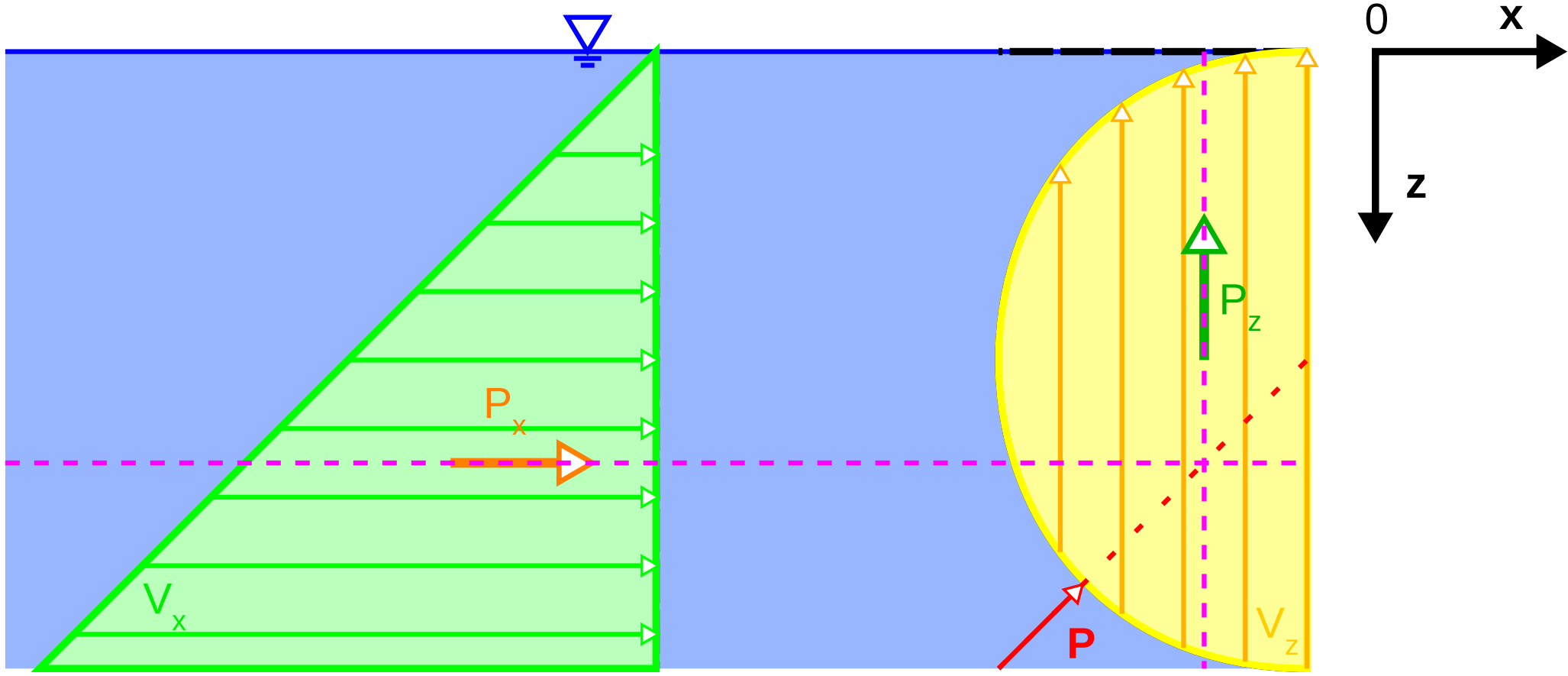

0
x
z
P_x
V_x


0
x
z
P_x
V_x
P_z
V_z


0
x
z
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x
V
x
P
z
V
z

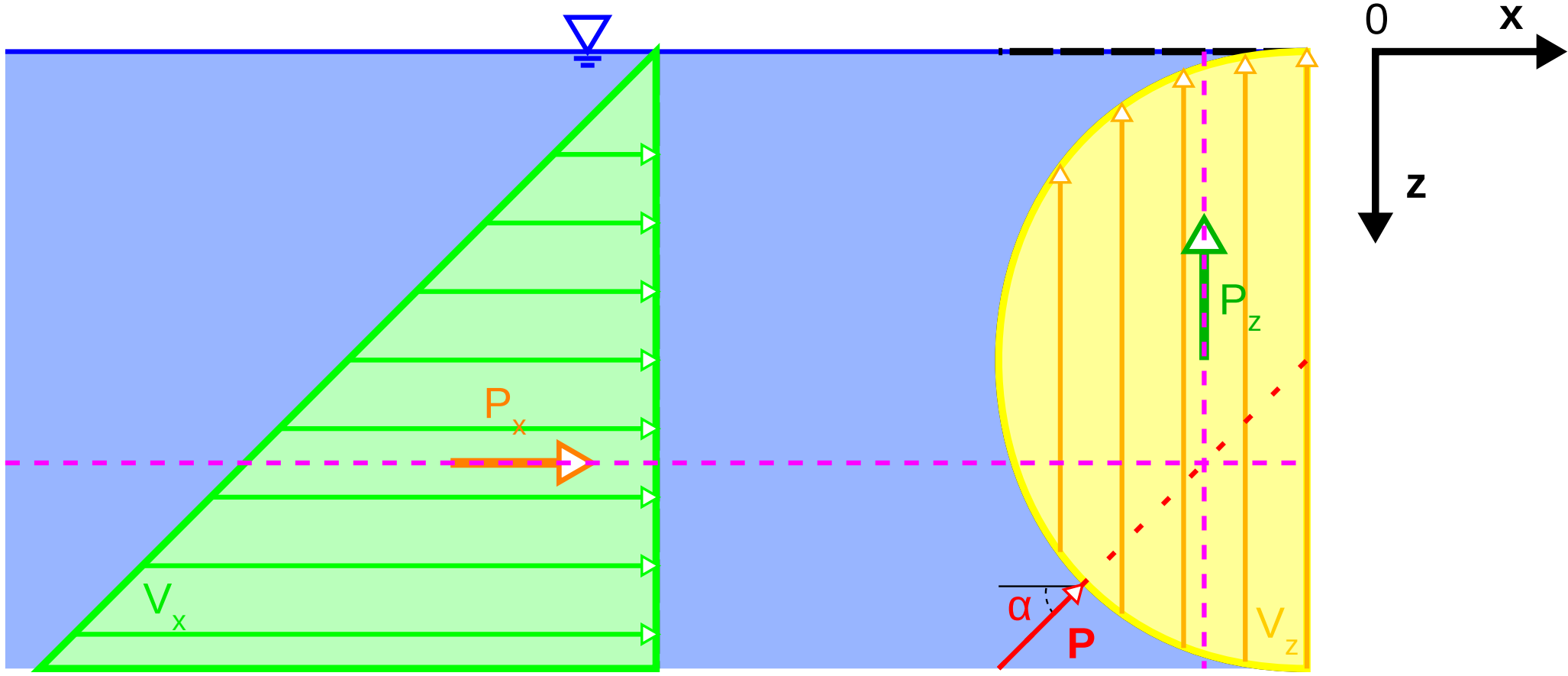

0
x
z
P_x
V_x
P_z
V_z

$$P = \sqrt{P_x^2 + P_x^2}$$



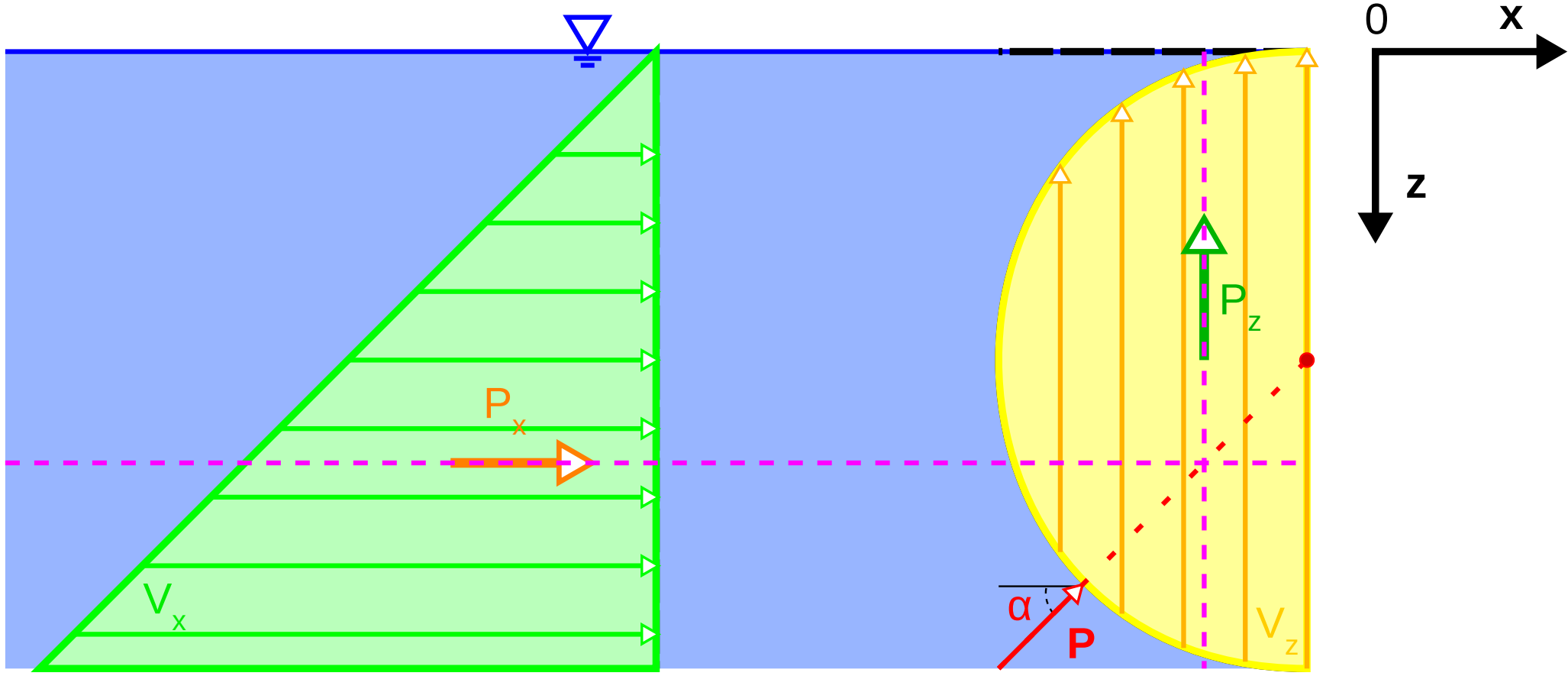
$$P = \sqrt{P_x^2 + P_x^2}$$

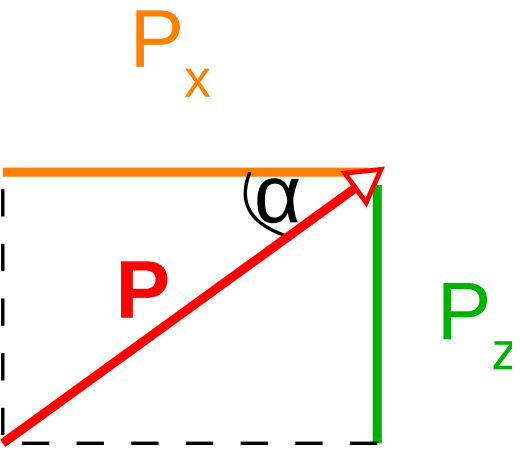
$$\alpha = \arctan \frac{P_z}{P_x}$$



$$P = \sqrt{P_x^2 + P_x^2}$$

$$\alpha = \arctan \frac{P_z}{P_x}$$





Wypadkowa parcia

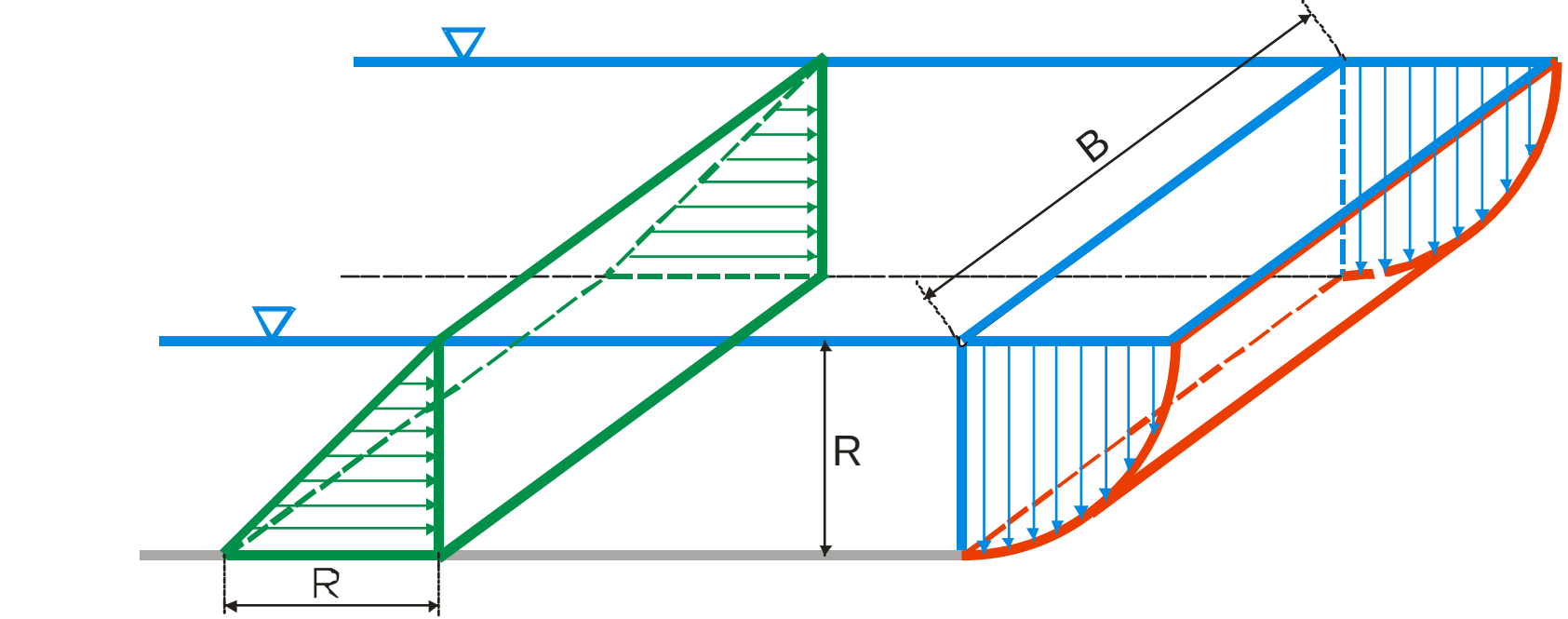
Długość wektora

$$P = \sqrt{P_x^2 + P_x^2}$$

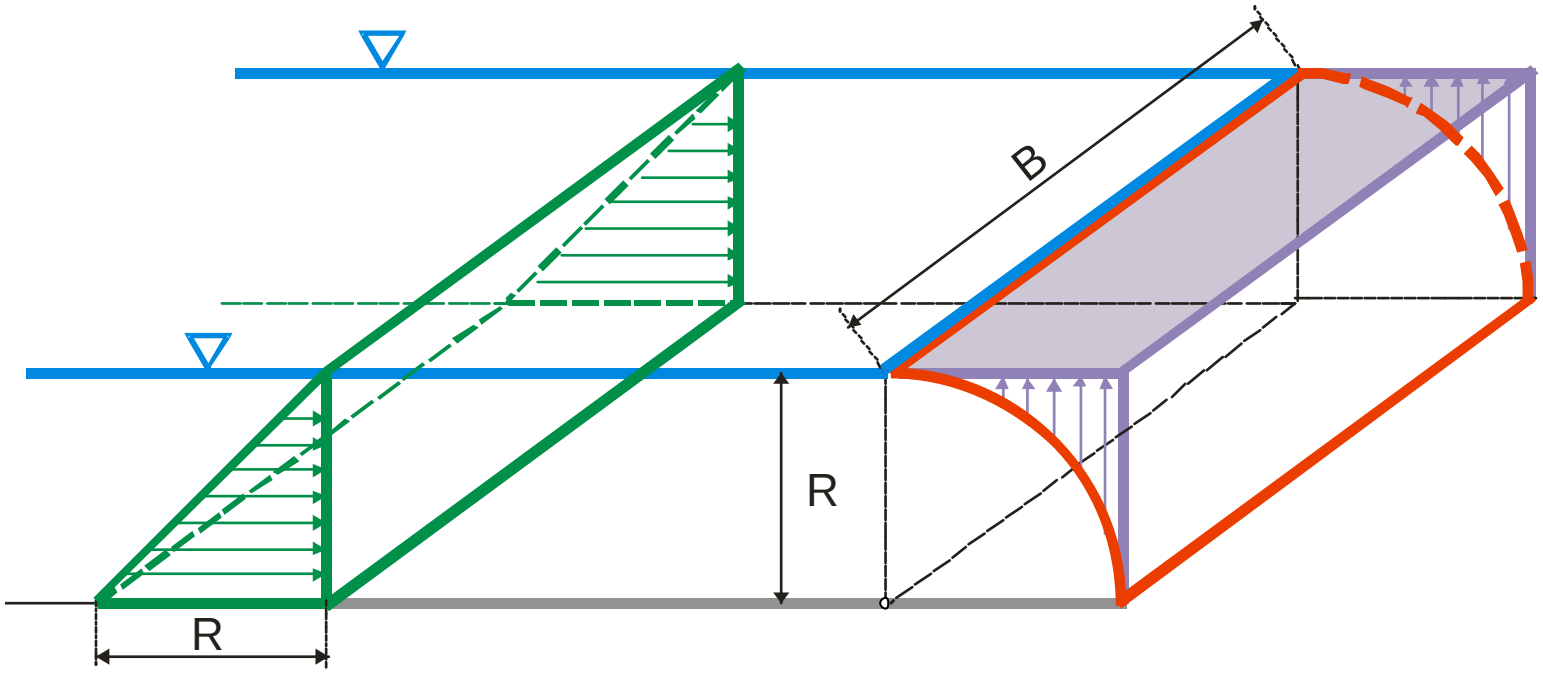
Kąt nachylenia od poziomu

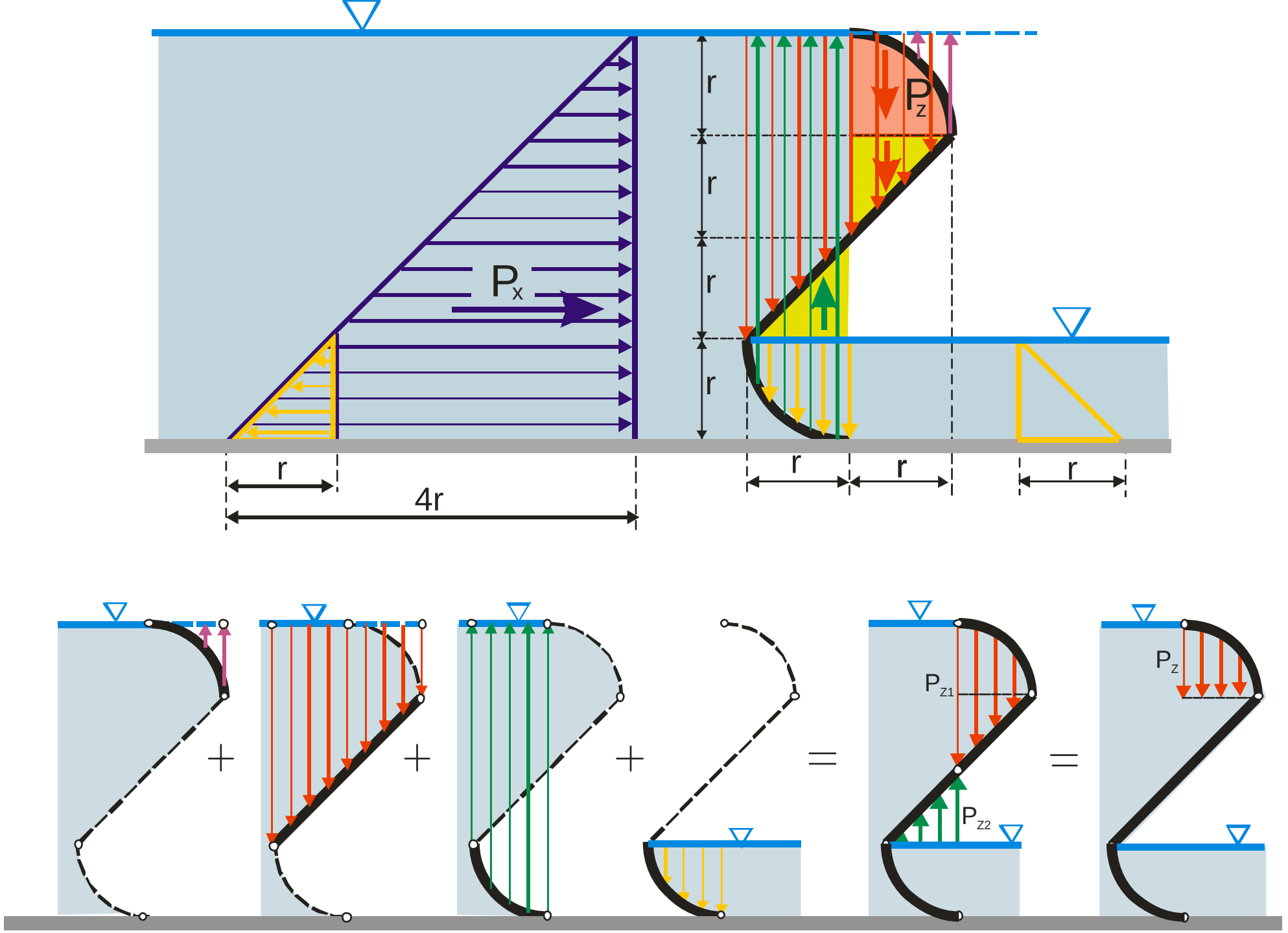
$$\alpha = \arctan \frac{P_z}{P_x}$$

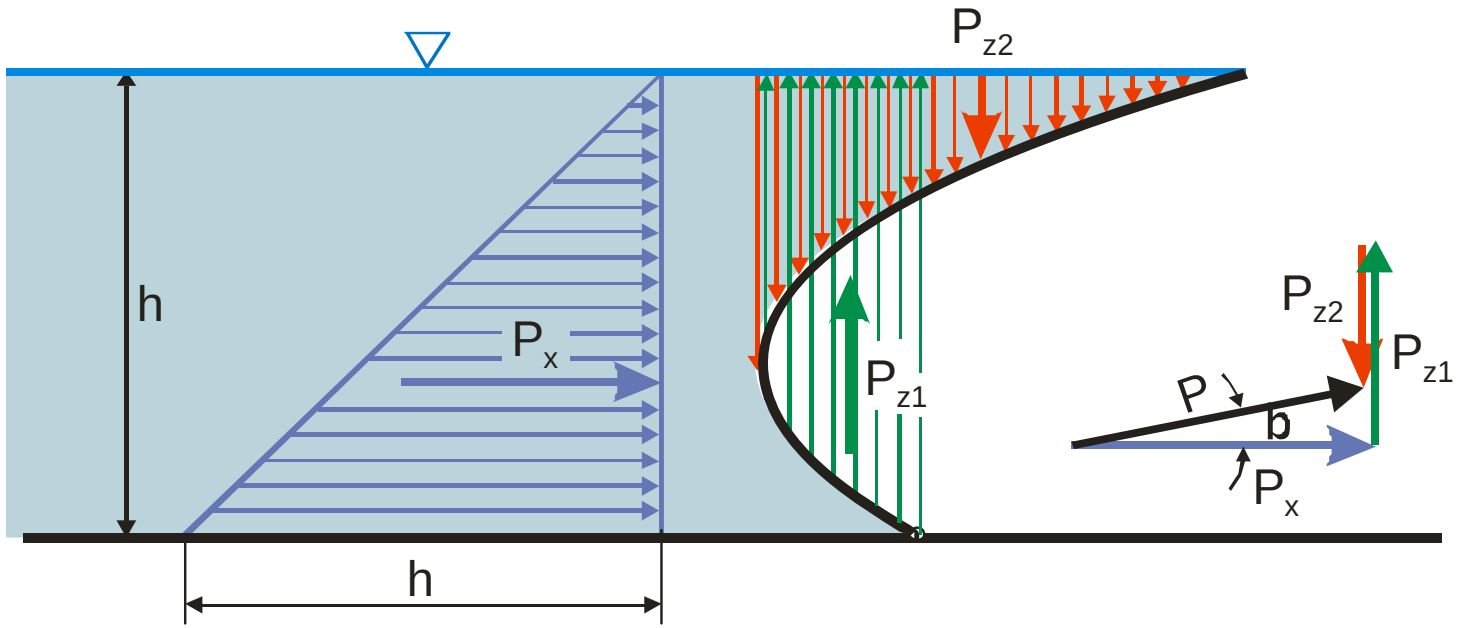
Przykłady

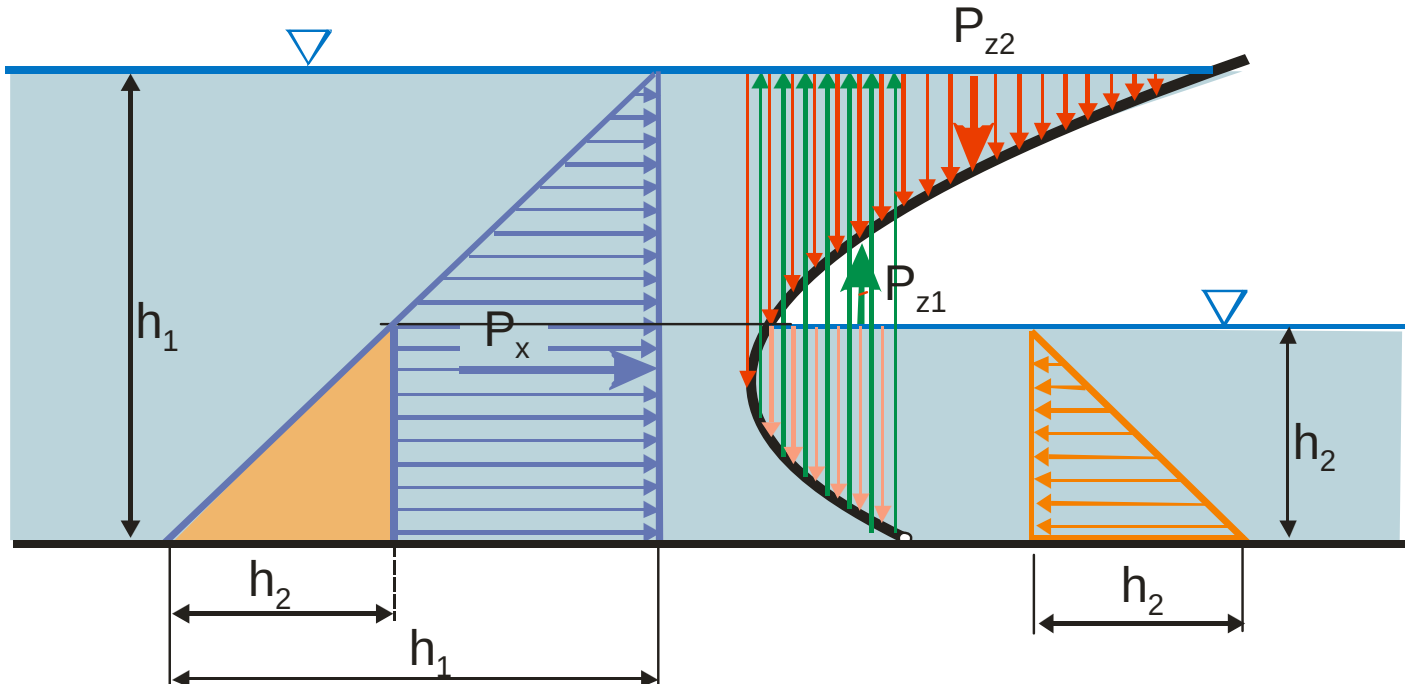


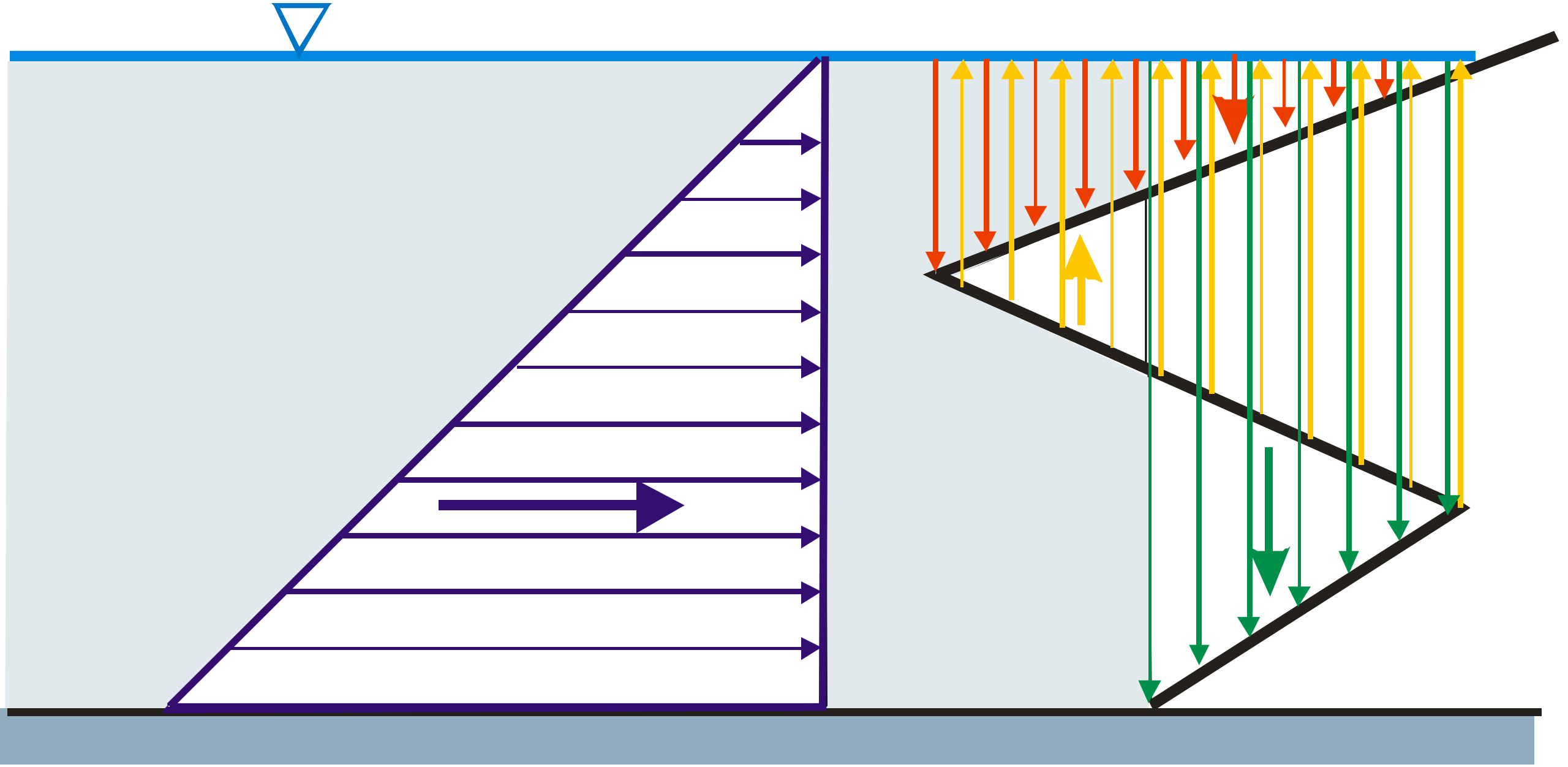
Z

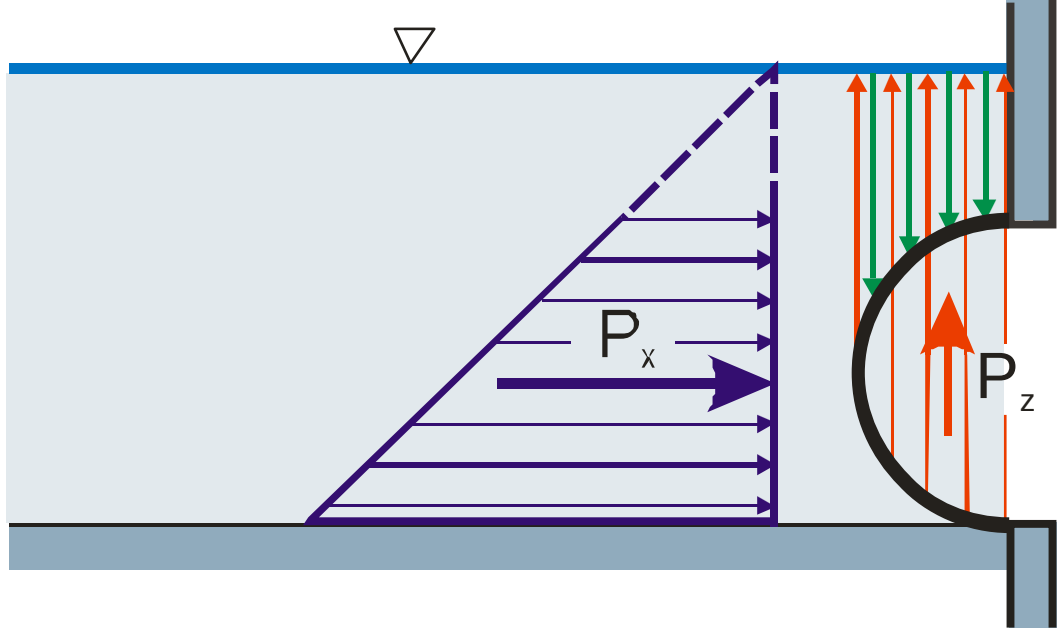


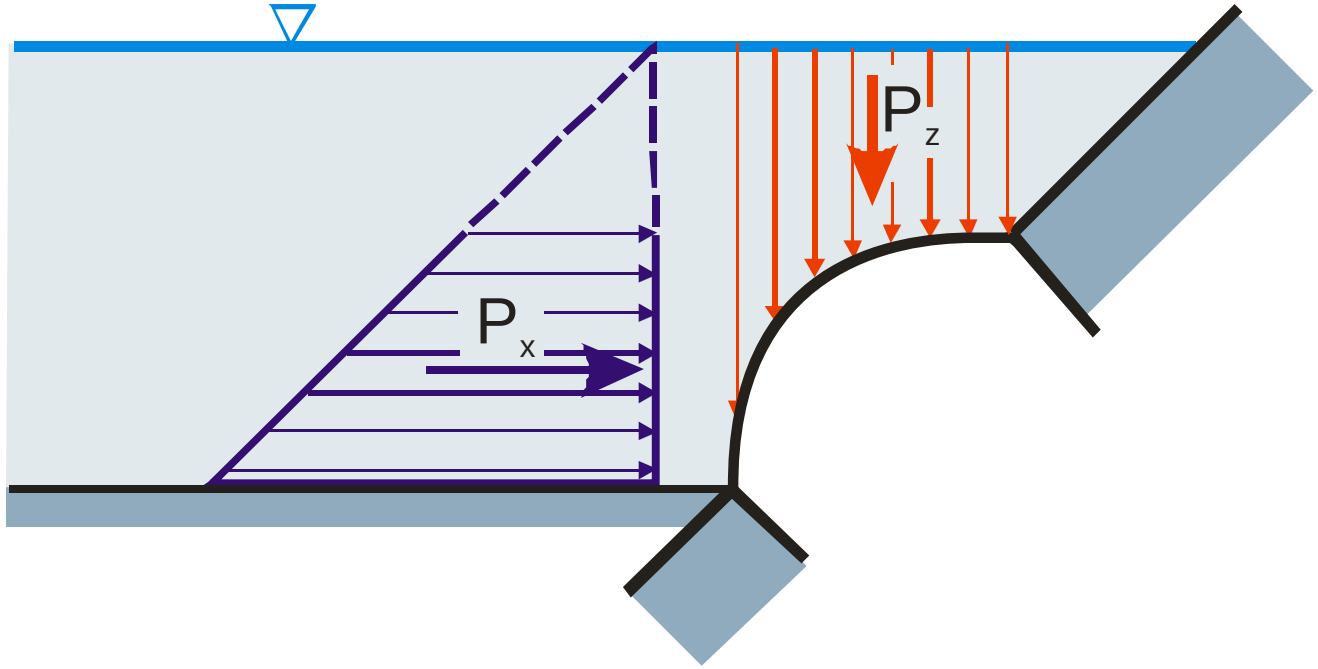

Px
Pz
r
r
r
r
r
r
r
4r
+
+
+
=
=
Pz1
Pz2
Pz


P z2
h
P x
P z1
P z2
P z1
P
b
P x
h

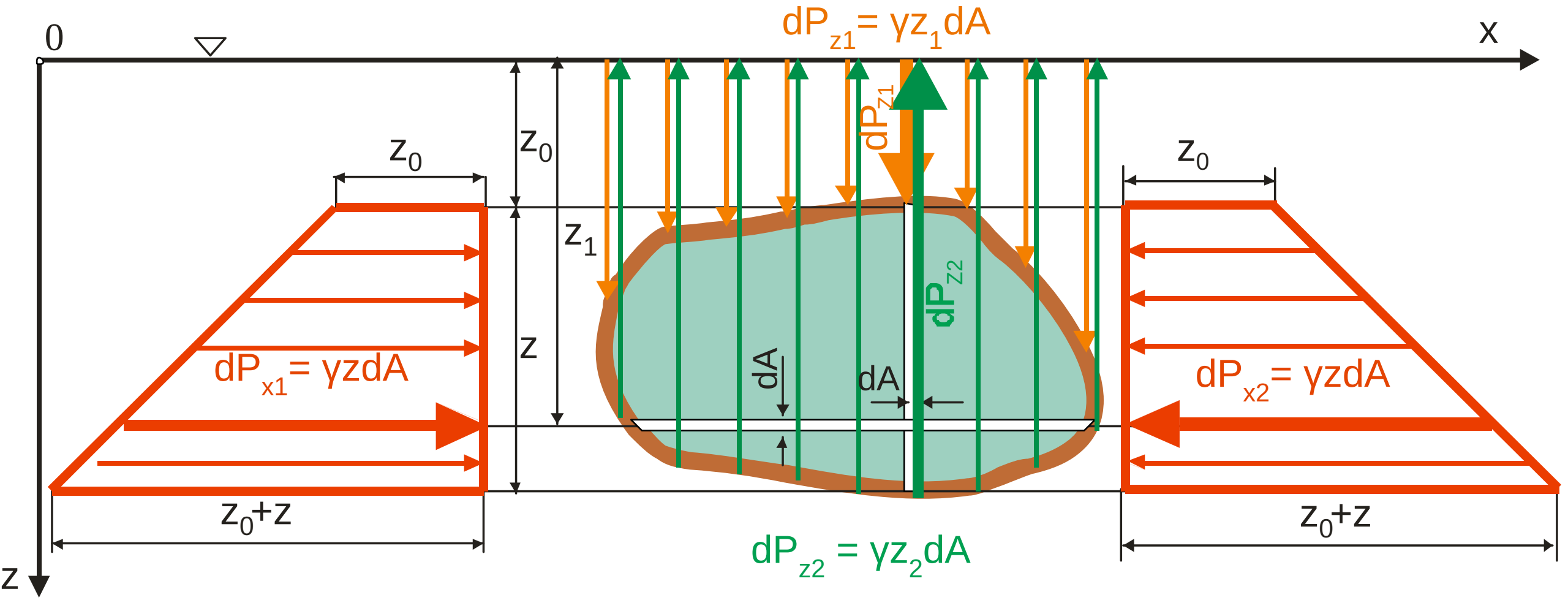

P z2
h 1
P x
P z1
h 2
h 2
h 2
h 1




P_x
P_z

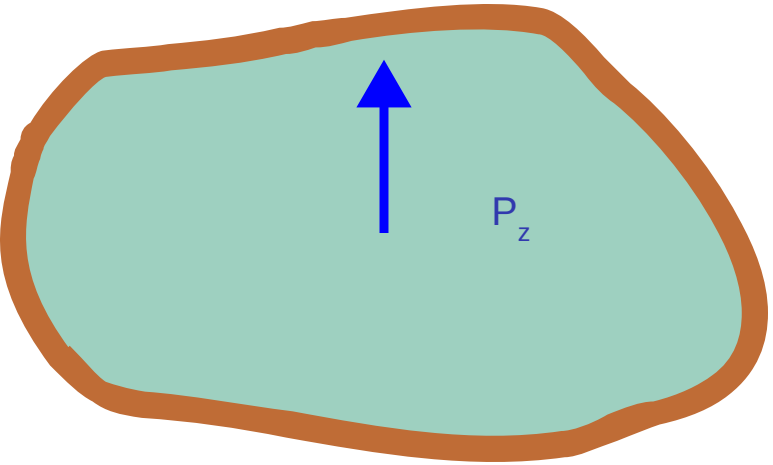

P_x
P_z

Wypór hydrostatyczny



$$dP_x = dPx_1 + (-dPx_2) = \gamma z_1 dA - \gamma z_1 dA = 0$$

$$dP_z = dPz_1 + (-dPz_2) = \gamma z_0 dA - \gamma(z_0 + z)dA = -\gamma z dA$$



- Na ciało zanurzone w cieczy działa tylko pionowa składowa parcia skierowana ku górze

$$P_z = \gamma \int zdA = \gamma V$$

- $\int zdA = \gamma V$ *to objętość zanurzonego ciała*
- Wypór hydrostatyczny jest siłą powierzchniową skierowaną pionowo do góry, równą iloczynowi objętości zanurzonego ciała i ciężaru objętościowego cieczy.
- **Prawo Archimedesa**: Ciało zanurzone w cieczy traci pozornie na wadze tyle, ile waży wyparta przez nie ciecz